SPECIFICATION COMPLIANCE- PIPE INSULATION		
SL.NO	**SPECIFICATION**	**Compliance**
15	**INSULATION**	
15.1	**Scope**	
	The scopeofthissection comprisesthesupplyand application of insulation conforming to these specifications. The insulation material shall be Closed Cell Elastomeric Nitrile Rubber / Polyethylene Foam / EPDM	
15.2	**Material**	
	Thermal insulationmaterialfor Duct & Pipe insulation shallbewith factory laminated black fiber glass cloth closed cell Elastomeric UV resistant. Thermal conductivity as per BS 874 part 2 – 86 (DIN 52613 52612) /DIN EN 12667 / EN ISO8497 ofthe insulation material shall not exceed 0.038 W/moKor0.212 BTU / (Hr-ft2-oF/inch) at an average temperature of 30oC.Density of the nitrile rubber shall be 40-60 Kg/m3, for EPDM shall be 40-60 Kg/m3& for polyethylene material it shall be 25-30 Kg/m3. The product shall have temperature range of –40 oC to 105oC. The insulation materialshall befire rated for Class 0 as per BS 476 Part 6 : 1989 for fire propagation test and for Class 1 as per BS 476 Part 7, 1987 for surface spread of flame test.Water vapour permeability shall be not less than 0.024 per inch (2.48 x 10-13 Kg/m.s.Pa i.e. µ>7000: Water vapour diffusion resistance) as per DIN 53122 part 2, DIN 52615 / EN 12086 & EN13469	**Complied**
15.4	**Piping Insulation**	
	All chilled water, refrigerant, and condensate drain piping shall be insulated in the manner specified here in. Before applying insulation,all pipe shallbe brushed and cleaned.All MS pipes shall be provided with a coat of zinc chromate primer. Thermal insulation shall be applied as follows or as specified in drawings or schedule of quantity: Piping Insulation thickness shall be as follows;	

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<tr><th>Pipe nominal bore</th><th>Thk. for non-coastal places</th><th>Thk. for coastal places</th></tr>
<tr><td>15 mm – 25mm</td><td>19 mm</td><td>25mm</td></tr>
<tr><td>32 mm – 80 mm</td><td>25 mm</td><td>32 mm</td></tr>
<tr><td>100 mm – 400 mm</td><td>32 mm</td><td>38 mm</td></tr>
<tr><td>Above 400 mm</td><td>45 mm</td><td>45 mm</td></tr>
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<td>25mm - Single Layer (25mm Nitrile with GC)
32mm - Single Layer (32mm Nitrile with GC)
38mm - Double Layer (25mm Plain + 13mm with GC)
45mm - Double Layer (32mm Plain + 13mm with GC)</td>
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<td>Insulating material in tube form (minimum upto 100 dia pipes) shall be sleeved on the pipes.On piping, slit opened tube from insulating material shall be placed over the pipe and adhesive shall be applied as suggested by the manufacturer.Adhesive must be allowed to tack dry and then press surface firmly together starting from butt end and working towards centre.Wherever flat sheets shall be used it shall be with self-adhesive and cut out in correct dimension using correct tools.Scissors or Hacksaw-blade shall not be allowed. All longitudinal and transverse joints shall be sealed by providing 50 mm wide fiber glass cloth laminated tape as per manufacturer recommendations.The adhesive shall be strictly as recommended by the manufacturer. The insulation shall be continuous over the entire run of piping, fittings and valves.All valves, fittings, joints, strainers etc.in chilled water piping shall be insulated to the same thickness as specified for the main run of piping and application shall be same as above.Valves bonnet, yokes and spindles shall be insulated in such a manner as not to cause damage to insulation when the valve is used or serviced.</td>
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<td>Direct contact between pipe and hanger shall be avoided. Hangers shall pass outside the saddle. Manufacturer shall supply PUF saddles with pre-laminated insulation sheet of both side (PUF saddle sandwich between insulation material on both side) so that the insulation material is joint with insulation material on both side (only for Nitrile & EPDM) and the weight of pipe is transferred to the PUF saddle in the center.</td>
<td>Noted</td>
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<td>Manufacturer’s installation manual shall be submitted and followed for full compliance.All insulation work shall be carried out by skilled workmen specially trained and certified by manufacturer in this kind of work.All insulated pipes shall be labeled (S.R. or R.R.) and provided with 300 mm wide band of paint along circumference at every 1200 mm for colour coding.Direction of fluid shall also be marked.Un-insulated MS pipes shall be painted throughout and direction of fluid marked.All painting shall be as per relevant BIS codes.</td>
<td>Noted</td>
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<td>15.5</td>
<td>Protective Coating / Vapour Barrier Over Insulation
All ducts and pipes (On the roof / outside) exposed to UV rays shall be covered with two coats of UV paint / epoxy.</td>
<td>Noted</td>
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Inner Engineering Products & Systems Pvt. Ltd.